

Q11 Prove Fermat's Little Theorem and use it to compute $a^{p-1} \bmod p$ for given values of $a \neq 0$, $p=13$. Then, discuss how this theorem is useful in cryptographic algorithms like RSA.

Ans:

Fermat's Little Theorem: If p is a prime number and a is an integer not divisible by p , then, $a^{p-1} \equiv 1 \bmod p$.

Proof:

Let a be an integer such that $\gcd(a, p) = 1$ and p is prime.

Consider the set of integers modulo p (excluding 0):

$$S = \{1, 2, 3, \dots, p-1\}$$

Multiplying each element of S by a modulo p gives another set:

$$S' = \{a \cdot 1, a \cdot 2, \dots, a \cdot (p-1)\} \bmod p$$

Since a is coprime with p , multiplication by a is a bijection (i.e., no duplicates and all residue appear).
So, $S' \equiv S \bmod p$

Now taking the product of all elements in S and S' :

$$a^{p-1} \cdot (1 \cdot 2 \cdot \dots \cdot (p-1)) \equiv (1 \cdot 2 \cdot \dots \cdot (p-1)) \bmod p$$

We can cancel factorial from both sides (since it's nonzero mod p), giving
 $a^{p-1} \equiv 1 \bmod p$ (proved)

* Using the theorem to compute $a^{p-1} \mod p$

Given that,
 $a=7$ and $p=13$

Check: $p=13$ is prime

$$\therefore \gcd(7, 13) = 1$$

$$\text{Then, } a^{p-1} = 7^{12}$$

By the Fermat's Little theorem:

$$7^{12} \equiv 1 \mod 13$$

$$\text{So, } 7^{12} \mod 13 = 1$$

* Use in Cryptography (RSA):

Fermat's theorem helps in:

(*) Modular exponentiation (used in encryption/decryption)

(*) Primality testing (to generate large prime numbers)

(*) Forms the base of Euler's theorem, which is used in RSA encryption.

Q2// Euler's Totient Function: compute $\phi(n)$
for $n=35, 45, 100$. Prove that if a and n are
coprime, then $a^{\phi(n)} \equiv 1 \pmod n$.

Ans:

Euler's Totient Function: $\phi(n)$

Euler's Totient function $\phi(n)$ counts the number
of integers from 1 to n that are coprime to
 n (i.e., $\gcd(k, n) = 1$).

Formula for $\phi(n)$:

If the prime factorization of n is

$$n = p_1^{k_1} p_2^{k_2} p_3^{k_3} \dots p_m^{k_m}$$

$$\text{Then, } \phi(n) = n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_m}\right)$$

Compute the $\phi(n)$ for Given values:

(i) For $n=35$

Prime factorization: $35 = 5 \cdot 7$

$$\phi(35) = 35 \cdot \left(1 - \frac{1}{5}\right) \cdot \left(1 - \frac{1}{7}\right) = 35 \cdot \frac{4}{5} \cdot \frac{6}{7} = 24$$

(ii) For $n=45$

Prime factorization: $45 = 3^2 \cdot 5$

$$\phi(45) = 45 \cdot \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{5}\right) = 45 \cdot \frac{2}{3} \cdot \frac{4}{5} = 24$$

(iii) For $n=100$

Prime factorization: $100 = 2^2 \cdot 5^2$

$$\phi(100) = 100 \cdot \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{5}\right) = 100 \cdot \frac{1}{2} \cdot \frac{4}{5} = 40$$

Euler's Theorem

If $\gcd(a, n) = 1$, then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Proof:

Let $\gcd(a, n) = 1$

Let the set of numbers less than n and coprime to n be:

$$R = \{r_1, r_2, \dots, r_{\phi(n)}\}$$

All r_i 's satisfy $\gcd(r_i, n) = 1$

Now consider:

$$a \cdot r_1, a \cdot r_2, \dots, a \cdot r_{\phi(n)} \pmod{n}$$

Since $\gcd(a, n) = 1$, multiplying by a permutes the elements in R (no duplicates)

So:

$$a \cdot r_1 \cdot a \cdot r_2 \cdot \dots \cdot a \cdot r_{\phi(n)} \equiv r_1 \cdot r_2 \cdot \dots \cdot r_{\phi(n)} \pmod{n}$$

Left side becomes:

$$a^{\phi(n)} \cdot (r_1 \cdot r_2 \cdot \dots \cdot r_{\phi(n)})$$

$$\text{so, } a^{\phi(n)} \cdot R \equiv R \pmod{n}$$

canceling R (since it's not divisible by n)

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

(Proved)

Q311 Solve the system of congruences using the Chinese Remainder Theorem and prove that x is congruent to 11 on mod $N = 3 \times 4 \times 5 = 60$.

$$x \equiv 2 \pmod{3} \quad x \equiv 3 \pmod{4} \quad x \equiv 1 \pmod{5}$$

Ans:

Given that,

$$x \equiv 2 \pmod{3} \quad x \equiv 3 \pmod{4} \quad x \equiv 1 \pmod{5}$$

We are given the system of congruences:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{4}$$

$$x \equiv 1 \pmod{5}$$

We are to solve it using the Chinese Remainder Theorem (CRT) and prove that the solution is:

$$x \equiv 11 \pmod{60}$$

Define the modulus values:

$$\text{Let, } m_1 = 3$$

$$m_2 = 4$$

$$m_3 = 5$$

These moduli are pairwise coprime (i.e., any two have $\text{gcd} = 1$), so CRT applies.

Total modulus:

$$M = m_1 \cdot m_2 \cdot m_3 = 3 \cdot 4 \cdot 5 = 60$$

Now, calculate $M_i = \frac{M}{m_i}$

$$M_1 = \frac{60}{3} = 20$$

$$M_2 = \frac{60}{4} = 15$$

$$M_3 = \frac{60}{5} = 12$$

Find inverse y_i such that:

$$m_i y_i \equiv 1 \pmod{m_j}$$

$$\times 20y_1 \equiv 1 \pmod{3} \Rightarrow y_1 = 2 \text{ (since } 20 \cdot 2 = 40 \equiv 1 \pmod{3})$$

$$\times 15y_2 \equiv 1 \pmod{4} \Rightarrow y_2 = 3 \text{ (since } 15 \cdot 3 = 45 \equiv 1 \pmod{4})$$

$$\times 12y_3 \equiv 1 \pmod{5} \Rightarrow y_3 = 3 \text{ (since } 12 \cdot 3 = 36 \equiv 1 \pmod{5})$$

Now Construct solution:

The solution is:

$$x \equiv a_1 m_1 y_1 + a_2 m_2 y_2 + a_3 m_3 y_3 \pmod{m}$$

$$\text{where, } a_1 = 2, a_2 = 3, a_3 = 1$$

$$\text{so, } x \equiv 2 \cdot 20 \cdot 2 + 3 \cdot 15 \cdot 3 + 1 \cdot 12 \cdot 3 \pmod{60}$$

$$x \equiv 80 + 135 + 36 = 251 \pmod{60}$$

Now reduce,

$$x \equiv 251 \pmod{60} \Rightarrow x \equiv 11 \pmod{60}$$

$$\therefore x \equiv 11 \pmod{60}$$

$$\begin{aligned} 6 &= 2 \cdot 3 \\ 12 &= 2^2 \cdot 3 \\ 15 &= 3 \cdot 5 \end{aligned}$$

$$M = m_1 m_2 m_3 = 2 \cdot 4 \cdot 3 = 24 = 60$$

$$\frac{M}{m_i} = M'$$

$$M' = \frac{60}{2} = 30$$

$$M' = \frac{60}{4} = 15$$

$$M' = \frac{60}{3} = 20$$

Ques Find whether 561 is a Carmichael number by checking its divisibility and Fermat's test.

Ans.

A Carmichael number is a composite number n such that:

$a^{n-1} \equiv 1 \pmod{n}$ for all integers a that are coprime to n .

check if 561 is prime:

$$561 = 3 \times 11 \times 17$$

So, 561 is composite.

A number n is a Carmichael number if:

(i) n is composite

(ii) n is square-free

(iii) For each prime divisor p of n , $p-1$ divides $n-1$.

Let's check:

$$* n = 561$$

$$* \text{Prime divisor } p_1 = 3, p_2 = 11, p_3 = 17$$

$$* n-1 = 560$$

Now check divisibility:

$$* 3-1 = 2, \text{ and } 2 \mid 560$$

$$* 11-1 = 10, \text{ and } 10 \mid 560$$

$$* 17-1 = 16, \text{ and } 16 \mid 560$$

So, all conditions satisfied.

Therefore, 561 is a Carmichael number.

Check Fermat's Test for a few values

Pick values a that are coprime to 561, and test:

$$a^{560} \pmod{561} = 1?$$

$$\text{Try } a = 2, 3, 10$$

$$* \gcd(2, 561) = 1, \text{ so, } 2^{560} \equiv 1 \pmod{561} \quad \square$$

$$* \gcd(3, 561) = 3 \neq 1 \rightarrow \text{not valid for Fermat's test}$$

$$* \gcd(10, 561) = 1, \text{ so, } 10^{560} \equiv 1 \pmod{561} \quad \square$$

So, Fermat's test does not detect 561 as composite.

Q51 Find a generator (primitive root) of the multiplicative group modulo 17.

Ans:

To find a primitive root modulo 17, we need a number g such that its powers:

$$g^1, g^2, \dots, g^{16} \pmod{17}$$

generate all values from 1 to 16, without repetition.

Since 17 is prime, the group $\mathbb{Z}_{17}^*$ has order 16, so we check:

$$g^{16/d} \not\equiv 1 \pmod{17} \text{ for } d = 2, 4, 8$$

Try $g=3$:

$$3^8 \equiv 16 \pmod{17} \neq 1$$

$$3^4 \equiv 13 \pmod{17} \neq 1$$

$$3^2 \equiv 9 \pmod{17} \neq 1$$

So, 3 passes all checks.

So, 3 is a primitive root modulo 17.

Q.11 Solve the Discrete Logarithm Problem: Find x such that $3^x \equiv 13 \pmod{17}$.

Ans:

Find x such that $3^x \equiv 13 \pmod{17}$

* Brute-force power of 3 modulo 17:

Let's compute successive powers of 3 modulo 17:

$3^x \pmod{17}$

x	$3^x \pmod{17}$
1	3
2	9
3	10
4	13 ✓
5	5
6	15
7	11
8	16
9	14
10	8
11	7
12	4
13	12
14	2
15	6
16	1

From the table:

we see that: $3^4 \equiv 13 \pmod{17}$

$\therefore x = 4$

Q7] Discuss the role of the discrete logarithm in the Diffie-Hellman key Exchange.

Ans:

The role of the discrete logarithm in the Diffie-Hellman key Exchange:

(i) Goal: Diffie-Hellman lets two users share a secret secret key over a public channel.

(ii) Public values: A large prime p and a base g (Primitive root mod p).

(iii) Private keys:

* Alice picks secret a , sends $A = g^a \text{ mod } p$

* Bob picks secret b , sends $B = g^b \text{ mod } p$

(iv) Shared secret:

* Alice computes $B^a \text{ mod } p$, Bob computes $A^b \text{ mod } p$.

* Both get the same result: $g^{ab} \text{ mod } p$.

(v) Security: Based on the Discrete Logarithm Problem (DLP): hard to find a from $g^a \text{ mod } p$.

(vi) Conclusion: Without solving the DLP, an attacker can't find the shared key.

$$g^a \text{ mod } p = g^b \text{ mod } p \Rightarrow a = b$$

$$p = 23 \therefore$$

Q81 Compare and contrast the substitution cipher, Transposition cipher, and Playfair cipher in terms of encryption mechanism, key space, and vulnerability to frequency analysis. provide an example plaintext and show how each cipher transforms it.

Ans:

Comparison Table:

Cipher	Encryption Mechanism	key space	Frequency Analysis Vulnerability
Substitution	Replaces each letter with another fixed letter (monoalphabetic)	$26! \approx 4 \times 10^{26}$	High - letter frequencies preserved.
Transposition	Rearranges positions of letters without changing them.	Depends on permutation length (e.g., $n!$ for n letters)	Low to medium - letter preserved but reordered.
Playfair	Encrypts digraphs (pairs of letters) using a 5×5 key square	$25! \approx 1.5 \times 10^{25}$ (since I=J)	Lower - frequency analysis harder due to digraph use

Example:

Plaintext: HELLO

(i) Substitution Cipher:

Key: A simple Caesar cipher (Shift by 3)

* H $\rightarrow$ K
 * E $\rightarrow$ H
 * L $\rightarrow$ O

* I $\rightarrow$ L
 * O $\rightarrow$ R

Cipher text: KHOOR.

Here each letter replaced by another.

(i) Transposition Cipher:

Key: Rearrange using block size 5, reverse order.

* Plaintext: HELLO

* Rearranged: OLLEH

Ciphertext: OLLEH

Each letter remain the same, only position change.

(ii) Playfair Cipher:

Key square (example using keyword MONARCHY):

M O N A R

C H Y B D

E F G I J K

L P Q S T

U V W X Z

Steps:

* Split plaintext into digraphs: HE LL OX (X is filler - for repeating L)

* Encrypt using Playfair rules:

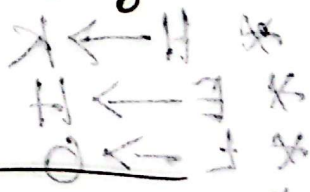
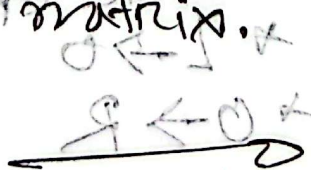
* HE $\rightarrow$ H & E in rectangle $\rightarrow$ C & F

* LL $\rightarrow$ same row $\rightarrow$ P & L

* OX $\rightarrow$ O & X in rectangle $\rightarrow$ Z & V

Ciphertext: C F P L Z V

Here encrypts by letter pairs using: Possible rules in a 5x5 matrix.



Q9// Given the Affine Cipher encryption function $E(x) = (ax+b) \bmod 26$, where $a=5$ and $b=8$.

(a) Encrypt the plaintext "Dept of ICT, MBSTU".

(b) Derive the decryption function and decrypt the ciphertext.

Ans:

Affine Cipher Formula:

Encryption: $E(x) = (ax+b) \bmod 26$

Decryption: $D(x) = a^{-1}(x-b) \bmod 26$

Given

(a) Ans:

Given, $a=5$, $b=8$

Plaintext: "Dept of ICT, MBSTU"

(i) Convert plaintext letters to numbers: We'll ignore non-alphabetic characters and convert letters. A-Z $\rightarrow$ 0-25

Letter	Number	Letter	Number
D	3	C	2
E	4	T	19
P	15	M	12
T	19	B	1
O	14	S	18
F	5	I	8
I	8	U	20

(i) Encrypt using $E(x) = (5x+8) \bmod 26$

Apply the formula:

x	$E(x) = (5x+8) \bmod 26$	x	$E(x) = (5x+8) \bmod 26$
3	$(5 \times 3 + 8) = 23 \rightarrow 23$	2	$(5 \times 2 + 8) = 18 \rightarrow 18$
4	$(5 \times 4 + 8) = 28 \rightarrow 2$	19	$(5 \times 19 + 8) = 103 \rightarrow 25$
15	$(5 \times 15 + 8) = 83 \rightarrow 5$	12	$(5 \times 12 + 8) = 68 \rightarrow 16$
19	$(5 \times 19 + 8) = 103 \rightarrow 25$	1	$(5 \times 1 + 8) = 13 \rightarrow 13$
14	$(5 \times 14 + 8) = 78 \rightarrow 0$	18	$(5 \times 18 + 8) = 98 \rightarrow 20$
5	$(5 \times 5 + 8) = 33 \rightarrow 7$	19	$(5 \times 19 + 8) = 103 \rightarrow 25$
8	$(5 \times 8 + 8) = 48 \rightarrow 22$	20	$(5 \times 20 + 8) = 108 \rightarrow 4$

(ii) Convert back to letters (0=A, ..., 25=Z)

Number	Letter	Number	Letter
23	X	18	S
2	C	25	Z
5	F	16	Q
25	Z	13	N
0	A	20	U
7	H	25	Z
22	W	4	E

$\therefore$ Ciphertext: "XCFZ AH WSZ, QNUZEF"

(b) Ans:

(i) Find modular inverse of $a=5 \bmod 26$.
we need a^{-1} such that,

$$5 \cdot a^{-1} \equiv 1 \bmod 26$$

Try values:

$$5 \times 21 = 105 \bmod 26 = 1$$

$$\text{So, } a^{-1} = 21$$

— P.T.O —

Decryption formula:

$$D(x) = 21(x-8) \bmod 26$$

Now decrypt the ciphertext: "XCFZ AH ^{WSZ} QNUZE"

Letter	X	$D(x) = 21(x-8) \bmod 26$	Plain
X	23	$21 \times (23) = 345 \rightarrow 3$	D
C	2	$21 \times (2) = -126 \rightarrow 9$	E
F	5	$21 \times (5) = -63 \rightarrow 15$	R
Z	25	$21 \times (25) = 357 \rightarrow 49$	T
A	0	$21 \times (0) = -168 \rightarrow 14$	O
H	7	$21 \times (7) = -21 \rightarrow 5$	F
W	22	$21 \times (22) = 298 \rightarrow 8$	I
S	18	$21 \times (18) = 210 \rightarrow 2$	C
Z	25	$21 \times (25) = 357 \rightarrow 49$	T
Q	16	$21 \times (16) = 168 \rightarrow 12$	M
N	13	$21 \times (13) = 105 \rightarrow 1$	B
U	20	$21 \times (20) = 252 \rightarrow 18$	S
Z	25	$21 \times (25) = 357 \rightarrow 49$	T
E	4	$21 \times (4) = -84 \rightarrow 20$	U

Decrypted Text: "DEPT OF ICT, MRSTU"

Q10] Design a simple novel cipher using a combination of substitution and permutation techniques. Describe its encryption and decryption processes. Then perform a basic cryptanalysis on your cipher to identify its potential vulnerabilities. You may use your own PRNG technique.

Ans:

Twist Cipher, encrypts plaintext using both substitution and permutation techniques. The plaintext is cleaned, uppercased, and split into 6 character blocks. A monoalphabetic substitution is applied using a shuffled alphabet generated from a 6 letter keyword (K1). Then a custom PRNG (Linear congruential Generator), seeded with integer $K2 + \text{block number}$, generates a permutation pattern for each block, used to shuffle character positions.

Encryption:

- (i) Substitute characters using the keyword-based shuffled alphabet.
- (ii) Permute characters in each block using the PRNG-generated index sequence.

P.T.O.

Decryption reverses permutation using the same PRNG and then reverses substitution using the inverse alphabet mapping.

Cryptanalysis:

- ⊗) Vulnerable to frequency analysis (monoalphabetic weakness).
- ⊗) Predictable PRNG can be reversed with known outputs.
- ⊗) Small key space and block size allow pattern detection.

To improve security, stronger PRNGs and polyalphabetic substitution can be used.

